

THE MUST KNOW SCALES

CHROMATIC

1	#1 b2	2	#2 b3	3	4	#4 b5	5	#5 b6	6	#6 b7	7	8
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MAJOR DIATONIC IONIAN MODE

1	2	3	4	5	6	7	8	I
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LYDIAN

1	2	3	#4	5	6	7	8	IV
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MIXOLYDIAN

1	2	3	4	5	6	b7	8	V
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MELODIC MINOR

1	2	b3	4	5	6	7	8
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← b6 b7 →

DORIAN

1	2	b3	4	5	6	b7	8	ii
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PHRYGIAN

1	b2	b3	4	5	b6	b7	8	iii
---	----	----	---	---	----	----	---	-----

AEOLIAN MODE NATURAL MINOR RELATIVE MINOR

1	2	b3	4	5	b6	b7	8	Vi
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HARMONIC MINOR

1	2	b3	4	5	b6	7	8
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MAJOR PENTATONIC

1	2	3	5	6	8	I
---	---	---	---	---	---	---

MINOR PENTATONIC

1	b3	4	5	b7	8	Vi
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WHOLE TONE

1	2	3	#4	#5	#6	8	+
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DIMINISHED

1	2	b3	4	b5	b6	bb7	7	8	O
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LOCRIAN

1	b2	b3	4	b5	b6	b7	8	Vii°
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MORE ABOUT SCALES

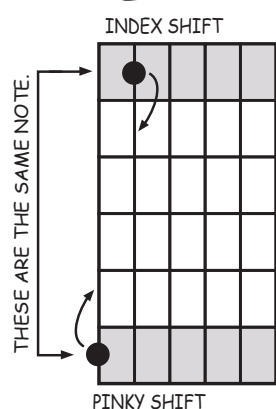
LESSON 15

You probably already know that one big thing about scales is that you have to ultimately know how to PLAY them on the neck of your guitar. All scales have MANY FINGERINGS because you can play any one in any of the 12 keys anywhere on the fret board. You don't need to know all the fingerings, but over your career, you owe it to yourself to soak up as many of them as you can. Your COLOR CODED FINGER CHARTS show 12 ways of playing each scale. These are the possible ways to play the scale within our 6 fret CHROMATIC "BOX" orientation. There are indeed other ways to play these scales as well (many involving SLIDES from one box to another, and you will pick these up over time from various sources). Of course, you could play any given scale on a SINGLE STRING simply by proceeding up the string using the correct pattern of half and whole steps for that scale. But how PRACTICAL a use of your fingers is that? What you're always trying to do is make the BEST and MOST LOGICAL use of all your fingers! That is the only way to attain real speed and control. Many players do not understand these 12 basic fingerings simply because they do not really understand their fret board in the way that YOU DO! You know that you can play all of the 12 notes, within a 5 fret range, for a distance of 2 and 1/3rd octaves anywhere on the neck! You understand how INDEX and PINKY SHIFTS work. What this means is that you should be able to play ANY SCALE... IN ANY KEY... ANYWHERE ON THE NECK! It may be a while before you see how AWESOME it is that you know this stuff and how far ahead of other players this puts you!

Many of these suggested guidelines for scale fingerings may at first seem more difficult or awkward than what you might come up with. But that is often simply because you haven't worked with them enough yet. Your fingers and ligaments haven't stretched enough yet to get the full benefit. You don't have enough EXPERIENCE to see the theoretical significance they hold. A classic example would be all those players who don't use their PINKY because it's such a weak finger. But it's only weak because you DON'T USE IT! You only have 4 fingers to begin with! Don't sacrifice your pinky just because you're too LAZY to get in there and work with it! That's 25% of your hand! It's a lot of work mastering these fingerings, but at the end comes TOTAL FREEDOM of a sort most players never even see is there! Your MODE JAM TAPE will make it more fun to practice your MAJOR DIATONIC MODES and your PENTATONIC SCALES. These are the most common scales and so the first ones you will want to get working on. But you will ultimately have to commit to a FOCUSED and DISCIPLINED practice regimen to master all your scales.

But first we must make sure that we know how to finger scales in general. We have to go back to what we learned about fingering CHROMATIC SCALES.

THE KEY TO UNDERSTANDING SCOTTY'S FAMOUS COLOR CODED FINGER CHARTS



Hopefully you all remember the significance of the 6 FRET CHROMATIC BOX. It takes 5 FRETS to control the 12 notes on the fret board, but that 5th note can be played with either a PINKY SHIFT on one string or an INDEX SHIFT on the next higher string. These can be any 6 frets anywhere on the neck. Remember that your G and B strings are tuned one fret closer together and that is bound to effect your fingering. As a rule, you'll find yourself using index shifts when going up the scale and pinky shifts when coming down but you will see that that isn't always the logical thing to do. Often, using INDEX REACHES when going both up and down help you avoid pinky shifts. When coming back onto the white frets after a FINGER REACH, make sure you use the correct finger that normally would have controlled that fret.

SO WHERE IS THE ROOT NOTE OF THE SCALE?

The root note can be any one of the 12 notes. That's how you tell which key you are playing the scale in. So the root can be under any finger on any fret on any string. It can even be a finger shift note on the shaded frets. That's what accounts for all these different finger patterns for the same scale. Note that as you go from box to box on any one of your charts, the RED DOT which is the ROOT NOTE, moves up one fret each time and so you are playing in the next higher key.

SO HOW DOES THE COLOR CODING SYSTEM WORK?

Remember back in grade school science class you learned the 7 COLORS OF THE SPECTRUM? They go

RED - ORANGE - YELLOW - GREEN - BLUE - INDIGO - VIOLET

Notice how most of our most common scales contain 7 NOTES (not including the octave). These are called the DEGREES OF THE SCALE. In our charts, each of the degrees of the scale is represented by each of the corresponding colors in the rainbow. So your ROOT NOTE is RED ...your 2nd is ORANGE ...your 3rd is YELLOW ...your 4th is GREEN ...your 5th is BLUE ...your 6th is INDIGO ...and your 7th is VIOLET. If for example, your 3rd is flatted (as in a minor scale), your YELLOW dot will have a flat sign on it etc. These color coded charts are so awesome because you can see all the degrees of the scale at a glance once you've memorized the color sequence. Compare that to your boring old black and white charts!

SO A MAJOR DIATONIC SCALE WOULD GO...



AND A NATURAL MINOR SCALE WOULD GO...



AND, SO INTERESTINGLY, THE TRIADS ARE IN PRIMARY COLORS!



STILL MORE ABOUT SCALES

LESSON 16

As we said before, the subject of scales is perhaps the most substantial subject in music theory. As always, we need to know these pitch patterns **4 WAYS ...THINK - READ - HEAR - SPEAK**. In this lesson, we look deeper into the **INTERVALS** that separate the notes in scales and begin to listen to them. Eventually you will want to **MEMORIZE** the sound of these scales so you can use them in composition and improvisation.

Many people are shocked to find out how many scales there are ...and all these weird names like **MIXOLYDIAN** and such. You might have taken quite a lot of music lessons and barely heard of them. Most of us have only heard of the **MAJOR SCALE**. Did you ever understand a **CHROMATIC SCALE** before you took this program? You might have heard that **PENTATONIC SCALES** and **BLUES SCALES** are big in Rock and Roll. But that's about it. How can Scotty suggest that all these other scales are equally important to know and even equally common in the songs we play? The first thing to know is this ...

THERE IS A SEPARATE SCALE FOR EACH CHORD IN THE SONG.

It is often suggested that an entire song is based on one scale (and certainly you can look at music that way but you might miss some important revelations). In Western Music, that fundamental scale is most often said to be the **MAJOR DIATONIC SCALE**. That's why it was the first one they tried to teach us back in grade school. A more modern way of teaching theory suggests that it is more useful to consider a separate scale for each chord in the song. How many chords are in the average pop song? Maybe between 3 and 6 ...sometimes more. This means that you are playing 3 to 6 of these scales in every tune you play! We most often see a mixture of **MAJOR** and **MINOR** chords in songs. What good is it to view the song as only containing one scale (say the **MAJOR SCALE**) if you have both major and minor chords? Doesn't it stand to reason that minor scales would work with minor chords? You will find that indeed there are very specific guidelines as to what scale goes with what chord. How do we know which scale works with which chord? This is a complex issue, but first consider this ...

THERE IS A CHORD EMBEDDED INSIDE EVERY SCALE
AND THE SCALE THAT GOES WITH THAT CHORD IS THE SCALE THAT HAS THAT CHORD INSIDE IT!
(Hopefully by this time you remember the number spellings for your major and minor chords)

MAJOR 1-3-5

MINOR 1-b3-5

In the video, we use **SILENT NIGHT** to demonstrate that even these strange scales like **LYDIAN** and **MIXOLYDIAN** show up in the most normal of songs if that is what the chord progression dictates. We will be looking at these relationships more in depth in our next few lessons on **MODES** and **HARMONIZATION**.

SO WHAT DO THESE SCALES SOUND LIKE?

CHROMATIC

Hopefully by now you are becoming familiar with the sound of the CHROMATIC SCALE. It is 13 notes separated by 12 half steps. You don't really play the complete scale very often in pop music but a lot of songs do contain brief chromatic passages. You need to know what a half step sounds like.

MAJOR SCALES

We've been listening to the MAJOR DIATONIC SCALE since we were kids. Now we are learning that there is MORE THAN ONE MAJOR SCALE. A major scale has a major chord inside it ...**1-3-5** and there are several scale patterns that fit that category. The #4 in LYDIAN may sound strange at first but you will notice it more and more as your ears become accustomed to it. Indeed it often "hides" in the melody to where you didn't even notice it there before. You will come to recognize the b7 in MIXOLYDIAN as being an INCREDIBLY common and important note in many contexts.

MINOR SCALES

There are MINOR CHORDS INSIDE MINOR SCALES ...**1-b3-5**. Most people would agree that the MINOR 3rd makes things sound SAD. Once again there are several slightly different minor scale patterns ...many again containing that b7. You may find it hard to believe that you will ever be able to pick up on their subtle differences by ear but indeed you can and must. Many pieces of music are written in MINOR KEYS where these scales replace the MAJOR SCALE as the central scale. In songs, minor scales interact with minor chords in very specific ways. Notice that the HARMONIC MINOR SCALE contains an AUGMENTED 2nd (1 and 1/2 step) interval.

PENTATONIC SCALES

PENTATONIC MEANS 5 NOTES. These scales contain two fewer notes per octave than their 7 note DIATONIC cousins. Therefore, these scales contain TWO AUGMENTED 2nd intervals which give them an open and "airy" sound that has attracted a lot of attention by various cultures throughout history. Some of you have heard that these are the ONLY scales you need to know to play Rock and Roll and Blues music but this is a terrible over simplification. Why only try to control 5 notes per octave when there is so much more? The MINOR PENTATONIC SCALE is often thought of as the foundation for the BLUES SCALE. It does contain the MINOR 3rd giving it a sad or "bluesy" feel. The b7 is also a huge note in blues. Try to remember what they sound like.

SYMMETRICAL SCALES

SYMMETRICAL SCALES GO WITH SYMMETRICAL CHORDS like AUGMENTED and DIMINISHED. A WHOLE TONE SCALE is all whole steps. A CHROMATIC SCALE is all half steps. A DIMINISHED SCALE goes whole-half-whole-half etc. Notice that these scales DO NOT CONTAIN NOTE 5 so they are NEITHER MAJOR OR MINOR! There are far fewer of these scales and chords in popular music partially because we are so used to that all important 5th that we don't quite know what to do when it isn't there. It is harder for us to memorize what these patterns sound like. Symmetrical chords are much more common in Jazz and Classical music, often snuck in between major and minor chords. LOCRIAN also has a b5 but notice that it is not symmetrical.

DIATONIC HARMONY A LA MODE

LESSON 17

WHY DO WE SEE THE SAME SETS OF CHORDS OVER AND OVER AGAIN IN DIFFERENT SONGS?

Those of you who have been playing for awhile will agree that there are hundreds of pop tunes that all share the same set of chords. One of the most common combinations in Pop Rock Music certainly is

G MAJOR - C MAJOR - D MAJOR - E MINOR - A MINOR

You might have played dozens of songs that have some or all of these chords in them. Why is that? Well you could just say that perhaps most song writers think that these chords sound good together. But there is a deeper level on which these chords are connected. Consider this statement ...

MOST SONGS CONTAIN ONLY 7 OF THE 12 NOTES. WHICH 7? THE 7 NOTES IN A MAJOR DIATONIC SCALE IN A GIVEN KEY.

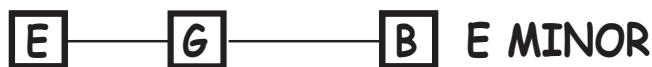
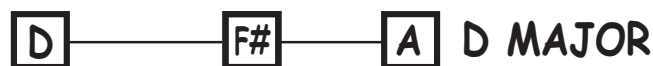
It's our good old friend the MAJOR DIATONIC SCALE again. So much of Western Music can be traced back to it. Now we find that the average song contains only the 7 notes in this scale. Not only is the MELODY of the song constructed out of these notes,

THE SONG CONTAINS ONLY THE CHORDS THAT CAN BE CONSTRUCTED OUT OF THOSE 7 NOTES!

Let's look at an example in the KEY OF G. The 7 notes in a G MAJOR SCALE are



WHAT CHORDS CAN BE MADE OUT OF THESE 7 NOTES?



AS YOU CAN SEE, THIS SET INCLUDES ALL THE CHORDS IN OUR EXAMPLE!

SO WHAT'S THIS GOT TO DO WITH MODES?

This is HUGE. You've learned a lot about PITCH PATTERNS so far in this program. We started with SILENCE and then we learned that all conventional patterns in music have a ROOT NOTE. We then talked about INTERVALS and suggested that the true nature of these patterns involves the SPACING of the notes. Now we're looking at the more complex note patterns like SCALES. You know all this stuff but you still don't see how this all comes together to EXPLAIN the songs we play. Well we are about to look at that now. We're going to do it in a rather unique way involving the concept of MODES. Many people have heard of modes. In fact many music students are taught a great deal about them without ever being shown their significance and relevance to the subject of how SCALES and MELODIES relate to CHORD PROGRESSIONS in conventional songs. The question is ...

HOW DO YOU KNOW WHAT SCALE GOES WITH WHAT CHORD?

Many of us who have done some JAMMING still remember when someone first told us that a single scale (often the MAJOR SCALE or the PENTATONIC SCALE) could be used to create melodies over a given set of chords. Wow! It seemed to make improvising so easy! We sound halfway like ERIC CLAPTON all ready! But if you play this way for long you begin to run out of ideas. You notice that all the notes in the scale sound pretty good but any given note will sound better or worse depending on which chord you play it over. What's going on? What do the guys on the radio know that you don't?

WOW! I CAN PLAY ONE SCALE OVER SEVERAL CHORDS!

NO! WHEN THE CHORD CHANGES, SO MUST THE SCALE CHANGE!

There is a separate scale for each chord. This makes sense doesn't it? How, for example, could a MAJOR SCALE work over both major and minor chords? Shouldn't we be playing minor scales over the minor chords? But it seems that the majority of people we talk to have never considered this possibility. They point out that playing one scale sounds pretty good and it would be too hard to switch back and forth from scale to scale every time the chord changes. Who ever heard of such a thing? But there is an answer.

THE SAME SET OF NOTES IS SEVERAL DIFFERENT SCALES. THE QUESTION IS WHICH NOTE IN THE SET IS "DO"?

THIS IS THE WHOLE THING ABOUT MODES.

MORE ABOUT MODES

LESSON 18

THE SAME SET OF NOTES IS SEVERAL DIFFERENT SCALES.

This is the whole concept of MODES. But what does this mean and why is it important for you to understand about them? Let's look at 2 octaves of a MAJOR DIATONIC SCALE. We recognize it by it's familiar pattern of intervals ... mostly WHOLE STEPS except between **3 and 4** and **7 and 8** or, in other words ...

WHOLE - WHOLE - HALF - WHOLE - WHOLE - WHOLE - HALF

But if you start on one of the other notes in the set, the interval pattern shifts and it is no longer accurate to view the new sequence of intervals as a Major Diatonic Scale. It is one of the MODES of the Major Diatonic Scale.

WHOLE	WHOLE	HALF	WHOLE	WHOLE	WHOLE	HALF	WHOLE	WHOLE	HALF	WHOLE	WHOLE	HALF
1	2	3 4	5	6	7 8	This 1st mode is the IONIAN ...the MAJOR DIATONIC SCALE itself. Notice that it is a MAJOR SCALE with a MAJOR CHORD (1-3-5) embedded inside it.						

1	2 b3	4	5	6 b7	8	Start on the 2nd note and your interval pattern changes. This 2nd mode, DORIAN is a MINOR SCALE containing a MINOR CHORD (1-b3-5)
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1 b2	b3	4	5 b6	b7	8	Start on the 3rd note and you're playing PHRYGIAN MODE . It is a MINOR SCALE. Notice the b2 and b6.
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1	2	3	#4 5	6	7 8	Start on the 4th note and you're playing LYDIAN MODE . It is a MAJOR SCALE with a #4.
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Start on the 5th note and you are playing the **MIXOLYDIAN**, a very important MAJOR SCALE with a b7.

1	2	3 4	5	6 b7	8
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This 6th mode is the **AEOLIAN**. Notice that it is a NATURAL MINOR SCALE. It is the very important RELATIVE MINOR mode.

1	2 b3	4	5 b6	b7	8
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Start on the 7th note and you're playing the **LOCRIAN MODE**. It is the only mode that is neither major or minor. It's b5 and b3 relates it to DIMINISHED.

1 b2	b3	4 b5	b6	b7	8
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THERE ARE 7 DIFFERENT SCALES THAT ALL CONTAIN THE SAME NOTES!

Let's look at our example again in the KEY OF G. Set your SLIDERULE up by putting G in the upper left window ...root note of the IONIAN MODE. There are 6 other scales that contain the same notes as a G MAJOR SCALE. They are ...

G	IONIAN	(G MAJOR DIATONIC)
A	DORIAN	
B	PHRYGIAN	
C	LYDIAN	
D	MIXOLYDIAN	
(E NATURAL MINOR)	E	AEOLIAN
	F#	LOCRIAN

Not only is this an interesting fact about how scales interrelate with one another, this also tells us everything about why we keep seeing those same **CHORD COMBINATIONS** over and over.

NOTICE THAT EACH OF THESE MODES HAS A CHORD EMBEDDED INSIDE IT.

In our example using the **KEY OF G**, we again see that familiar set of chords. Each chord is rooted on one of the **DEGREES OF THE MAJOR DIATONIC SCALE** and is represented by a **ROMAN NUMERAL**. Some of these chords are **MAJOR** and some are **MINOR** depending on whether the mode it is in has a **MAJOR 3rd** or a **MINOR 3rd** in it. These modes are always in the same order and retain their same major/minor orientation no matter which **KEY** you play them in so you absolutely have to memorize this set of relationships.

THIS ONE SYSTEM OF PITCH ORGANIZATION MAY EXPLAIN THE MELODIES AND CHORD PROGRESSIONS FOR 50-75% OF ALL THE SONGS YOU HAVE EVER HEARD!

I	G	2	B 4	D	6	7 8	G MAJOR is the TONIC CHORD in the key of G and can be found inside G IONIAN. But you're not playing this scale over the other chords in the song.
ii	A	2 C	4	E	6 b7	8	A MINOR is the ii CHORD in the key of G. It is embedded inside the Dorian Mode and A DORIAN is the scale you are playing to it.
iii	B b2	D	4	F# b6	b7	8	B MINOR is the iii CHORD. You'll be using B PHRYGIAN to jam with over this chord
IV	C	2	E	#4 G	6	7 8	C MAJOR is the IV CHORD. The 7 notes are functioning as C LYDIAN over this chord.
V	D	2	F# 4	A	6 b7	8	D MAJOR is the all important V CHORD and so MIXOLYDIAN is a common and most important scale.
Vi	E	2 G	4	B b6	b7	8	E MINOR is the Vi CHORD. It is the next most common chord after I-IV-V so AEOLIAN, the RELATIVE MINOR mode, is very common.
Vii	F# b2	A	4 C	b6	b7	8	F# DIMINISHED is the Vii CHORD. This chord has us playing LOCRIAN MODE over it. Note the b3 and b5 which make this scale a HALF DIMINISHED scale.

IF THESE SCALES ALL CONTAIN THE SAME NOTES AS A G MAJOR SCALE, WHAT'S THE HARM IN THINKING THAT YOU'RE PLAYING G MAJOR OVER ALL THESE CHORDS? WHY DO I NEED TO LEARN ALL THESE NEW SCALES WITH THESE WEIRD GREEK NAMES?

You see it's all about **FUNCTION - FUNCTION - FUNCTION!** Yes, these 7 notes will sound pretty good over these 7 chords but the more you listen, the more you begin to hear that each individual note sounds different depending on which chord you play it over. That is because each note changes **NUMERIC FUNCTION** with each chord change. Sometimes it is an **IMPORTANT** note like the 3 or the 5 and you will want to use it often ...hold it longer ...begin and end **PHRASES** with it. But over another chord, this same note might be functioning as a less important melody note like 4 or 6 and you will need to be more careful with it. Understanding these shifting relationships and being able to control them through **EAR TRAINING** is what separates the great improvisers from the "LICK BARFERS" and finger players. Yes it will take more work to learn to jam this way but the rewards are very real. You could learn to solo as well or better than many of your great guitar heroes!